

Ayan Bhattacharjee

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RESEARCH INTERESTS

- 📖 **Accretion-Ejection around Compact Objects:** X-Ray Spectra/Timing Studies, Smoothed Particle Hydrodynamics, Monte Carlo Simulations, Radiative Transfer
- 📖 **Dynamics of Jets:** Relativistic Hydrodynamics, Radio Emission Modelling, Hydrodynamic Instabilities

EMPLOYMENT

- | | |
|---|---|
| 🏢 Postdoctoral Researcher
Department of Physics and Astronomy, SNU | South Korea
Oct 2025 – Present |
| 🏢 Postdoctoral Researcher
Research Institute of Basic Sciences, SNU
NRF CREATIVE AND CHALLENGING (창의·도전) RESEARCH FELLOW | South Korea
Jan 2025 – May 2025 |
| 🏢 Postdoctoral Researcher
Department of Physics, UNIST
NRF CREATIVE AND CHALLENGING (창의·도전) RESEARCH FELLOW | South Korea
Jun 2022 – Dec 2024 |
| 🏢 Researcher
Center for High-Energy Astrophysics, UNIST | South Korea
Jul 2021 – Jun 2022 |
| 🏢 Visiting Researcher
Department of Astrophysics and Cosmology, SBNBCBS | India
Aug 2019 – Jul 2020 |
| 🏢 Senior Research Fellow
Department of Astrophysics and Cosmology, SBNBCBS | India
Aug 2016 – Jul 2019 |
| 🏢 Junior Research Fellow
Department of Astrophysics and Cosmology, SBNBCBS | India
Aug 2014 – Jul 2016 |

EDUCATION

- | | |
|--|-------------------------------------|
| 🎓 Ph.D. in Astrophysics
🏛️: S. N. BOSE NATIONAL CENTRE FOR BASIC SCIENCES
THESIS: Spectral And Timing Properties Of Black Holes And Neutron Stars In X-Ray Binaries Using Two-Component Advective Flow Solution
Advisor: Prof. Sandip K. Chakrabarti | India
Aug 2014 – Feb 2021 |
| 🎓 M.Sc. in Physical Sciences (Graduated 1st in Class of '14)
🏛️: S. N. BOSE NATIONAL CENTRE FOR BASIC SCIENCES
PROJECT: Parrondo's Paradox and the Brownian Ratchet
Project Supervisor: Prof. Punyabrata Pradhan | India
Aug 2012 – Jul 2014 |
| 🎓 B.Sc. in Physics (Graduated 2nd in Class of '12)
🏛️: WEST BENGAL STATE UNIVERSITY | India
2009 – 2012 |

REFEREED PUBLICATIONS

- 📖 **First Author:** 5 [2 MNRAS[†], 2 ApJ[†], 1 ASSP[†]]
- ✉ **Corresponding Author:** 5
- 📖 **Second Author:** 3 [1 ApJ[†], 1 RAA, 1 AdSpR]
- 📖 **Contributory Author:** 3 [1 ApJ, 1 ApSS, 1 AdSpR]
- ☰ **Complete List of Publications:** Google Scholar, NASA ADS, ORCID

PROFESSIONAL SERVICES

- 🔗 **Editor:** Two-Component Advective Flow (TCAF) XSPEC User Manual, 2024
- 👥 **Member:** Korea Numerical Astrophysics Group (KNAG), 2023-Present
- 👥 **Member:** Korean Astronomical Society (KAS), 2021-Present
- 👥 **Member:** Center for High-Energy Astrophysics (CHEA), UNIST, 2021-2022
- 👤 **Peer-Reviewer:** Research in Astronomy and Astrophysics (RAA), 2020-Present
- 👤 **Resource Personnel:** Two-Component Advective Flow (TCAF) XSPEC User Group, 2014-Present

SELECTED PUBLICATIONS

1. **A. Bhattacharjee**, I. Banerjee, A. Banerjee, D. Debnath, S. K. Chakrabarti, "The 2004 outburst of BHC H1743-322: analysis of spectral and timing properties using the TCAF solution", **MNRAS**, **466**, 1372-1381 (2016)
2. **A. Bhattacharjee**, S. K. Chakrabarti, "Monte Carlo Simulations of Thermal Comptonization Process in a Two Component Advective Flow around a Neutron Star.", **MNRAS**, **472**, 1361-1371 (2017)
3. **A. Bhattacharjee**, "Generalized Flows Around Neutron Stars", in Mukhopadhyay B., Sasmal S. (eds) *Exploring the Universe: From Near Space to Extra-Galactic*, **ASSP**, vol 53. Springer, Cham, 93-107 (2018)
4. **A. Bhattacharjee**, S. K. Chakrabarti, "Timing Properties of Shocked Accretion Flows around Neutron Stars in presence of cooling", **ApJ**, **873**, 119 (2019)
5. A. Banerjee, **A. Bhattacharjee**, D. Debnath, S. K. Chakrabarti, "Spectral Analysis of χ Class Data of GRS 1915+105 Using TCAF Solution", **RAA**, 20(12), 208 (2020)
6. A. Banerjee, **A. Bhattacharjee**, D. Chatterjee, D. Debnath, S. K. Chakrabarti, T. Katoch, & H. M. Antia, "Accretion Flow Properties of GRS 1915+105 During Its θ Class Using AstroSat Data", **ApJ**, 916(2), 68 (2021)
7. D. Chatterjee, D. Debnath, A. Jana, J. R. Shang, S. K. Chakrabarti, H. K. Chang, A. Banerjee, **A. Bhattacharjee**, K. Chatterjee, R. Bhowmik, S. K. Nath, "AstroSat observation of non-resonant type-C QPOs in MAXI J1535-571", **ApSS**, 366(8), 82 (2021)
8. S. Chowdhury, S. Sasmal, J. Brundell, S. Chakraborty, **A. Bhattacharjee**, & S. K. Chakrabarti, "Energetic electron precipitation during lightning activities over Indian landmass as observed from WWLLN and NOAA-15 satellite", **AdSR**, 68(10), 4205 (2021)
9. A. Banerjee, **A. Bhattacharjee**, D. Debnath, S. K. Chakrabarti, "Similarities and differences in accretion flow properties between GRS 1915+105 and IGR J17091-3624: A case study", **AdSR**, 69(7), 2930 (2022)
10. **A. Bhattacharjee**, J. Seo, D. Ryu, & H. Kang, "A Simulation Study of Low-Power Relativistic Jets: Flow Dynamics and Radio Morphology of FR-I Jets", **ApJ**, 976, 91 (2024)
11. A. Banerjee, A. Garg, D. Rawat, et al., "Contemporaneous X-ray and Optical Polarization of EHSP Blazar H 1426+428", **ApJ**, (in Press), arXiv:2504.12410, (2025)

GRANTS, FELLOWSHIPS AND ACHIEVEMENTS

- 🏆 **Creative and Challenging Research Grant**: "Simulation Study on Low-Powered FR-I Jets from Radio Galaxies", total budget of 210,000,000 KRW (\$169,195), National Research Foundation of Korea, 2022-2025
- 📺 **SERB-ITS Grant**: Presenting findings at FOXT, API, Amsterdam, DST, India, 2019
- 📺 **COSPAR Grant**: An €800 support for 42nd COSPAR Assembly, COSPAR Secretariat, Caltech, USA, 2018
- 🌟 **Eligibility for Lectureship/Assistant Professorship**: CSIR-UGC NET, India, 2015-2016
- 🌟 **Eligibility for Scientific Officer**: A 99.6 percentile in OCES/DGFS 2014, BARC, India, 2014
- 🌟 **Eligibility for Engineering M.Tech.**: All India Rank 172 in GATE, India, 2014
- 🏆 **Gold Medal**: 1st position in IPhD Programme (2012-2014), Dean (AP) & Director of SBNBCBS, DST, 2014
- 🎓 **PBIR Fellowship**: Scholarship for Post-B.Sc. Integrated-PhD Scholar, SBNBCBS, DST, India, 2012
- 🏆 **Gold Medal**: 1st position in B. Sc.(H) Physics, BRSN College, India, 2012
- 🏆 **INSPIRE (Scholarship for Higher Education)**: Top 1% in the 10th & 12th standard, DST, India, 2009

COMPUTATIONAL SKILLS

- PRIMARY: **Fortran** [Advanced], **Shell** [Advanced], **Mathematica** [Advanced]
- SECONDARY: **ROOT** [Advanced], **Python** [Proficient], **Matlab** [Intermediate], **C/C++** [Intermediate]
- 📊 DATA VISUALIZATION: **GNUplot** [Advanced], **SuperMongo** [Advanced], **IDL** [Advanced], **Grace** [Advanced], **ParaView** [Intermediate], **Grapher** [Intermediate], **Origin** [Intermediate]
- 🔑 VERSION CONTROL AND PARALLEL COMPUTING: **git** [Proficient], **OpenMP** [Intermediate], **MPI** [User Experience]
- 📄 WORD PROCESSING: **LaTeX** [Advanced], **Pages** [Advanced], **Hancom Office** [Proficient], **Markdown** [Proficient]
- 📁 REFERENCE MANAGEMENT: **BibTex** [Advanced], **Obsidian** [Advanced], **Zotero** [Advanced]
- 🖥️ OPERATING SYSTEMS: **Linux** [Advanced], **MacOS** [Advanced], **Windows** [Proficient]

DATA REDUCTION & ANALYSIS SKILLS:

- 📦 PACKAGES: **HEASOFT/FTools** [Advanced], **HEASOFT/Xanadu** [Xspec, Xronos; Advanced].
- 📦 MISSIONS: **RXTE/PCA** [Advanced], **RXTE/HEXTE** [Proficient], **AstroSat/LAXPC** [Intermediate].

SELECTED INVITED AND SOLICITED TALKS

SNU ASTRONOMY COLLOQUIA

SNU, Seoul, South Korea

🔊: *What is the Origin of Jets in Accreting Neutron Stars?*

April 17, 2025

A Unified Accretion-Ejection Mechanism for Compact Objects

Two Component Advective Flows (TCAF):

ICSP, Kolkata, India

Fitting Procedure and Results for Stellar and supermassive black holes

🔊: *X-ray Spectral fitting of BHXRBs by TCAF FITS file*

Sep 25, 2024

THE 2023 SEPTEMBER KNAG MEETING

KASI, Daejeon, South Korea

🔊: *A Simulation Study of Low-Power Relativistic Jets:*

Sep 15, 2023

Structures and Dynamics of FR-I Jets

THE 68TH GWNR WORKSHOP

APCTP, POSTECH, Pohang, South Korea

🔊: *Numerical Simulations of Accretion-Ejection around Compact Objects:*

Mar 15-16, 2023

What to include (and what not to)?

SELECTED TALKS FROM INTERNATIONAL CONFERENCES

THE 16TH EAST ASIAN VLBI WORKSHOP (EAVW)

Pyeongchang, South Korea

🔊: *A Simulation Study on Relativistic Jets:*

Nov 17 - 21, 2025

Impact of the Central kpc Region on Jets across Different Scales

THE 45TH COSPAR ASSEMBLY, SESSION E1.2

BEXCO, Busan, South Korea

🔊: *What is the Origin of Jets in Accreting Neutron Stars?*

Jul 13 - 21, 2024

A Unified Accretion-Ejection Mechanism for Compact Objects

THE 45TH COSPAR ASSEMBLY, SESSION E1.8

BEXCO, Busan, South Korea

🔊: *A Simulation Study on Relativistic Jets:*

Jul 13 - 21, 2024

Impact of the Central kpc Region on Jets across Different Scales

THE XXXIst IAU GA MEETING, FOCUS MEETING 1

BEXCO, Busan, South Korea

🔊: *A Simulation Study on the Morphological Dichotomy of FR-I and FR-II Jets*

Aug 2-11, 2022

THE 43RD COSPAR ASSEMBLY, SESSION E1.5

Online, Sydney, Australia

🔊: *What is the Origin of QPOs in Accreting Neutron Stars?*

Jan 28 - Feb 4, 2021

THE 43RD COSPAR ASSEMBLY, SESSION E1.8

Online, Sydney, Australia

🔊: *Can there be a Unified Spectral Model for Black Holes and Neutron Stars?*

Jan 28 - Feb 4, 2021

THE FUTURE OF X-RAY TIMING

API, Amsterdam, Netherlands

🔊: *Can a Two-Component paradigm explain the spectral and timing properties of neutron stars?*

Oct 22 - 25, 2019

THE 42ND COSPAR ASSEMBLY, SESSION E1.13

Caltech, Pasadena, CA, USA

🔊: *Formation of Two-Component Advective Flows around Neutron Stars and the Possibility of Super-Eddington Accretion Rates*

Jul 14 - 22, 2018

THE 42ND COSPAR ASSEMBLY, SESSION E1.10

Caltech, Pasadena, CA, USA

🔊: *Formation and Stability of Oscillating Shocks in Inviscid Advective Flows around Neutron Stars in Presence of Cooling using Smoothed Particle Hydrodynamics Simulations*

Jul 14 - 22, 2018

THE 42ND COSPAR ASSEMBLY, SESSION E1.4

Caltech, Pasadena, CA, USA

🔊: *The Formation of Two Component Advective Flow around Neutron Stars*

Jul 14 - 22, 2018

THE 15TH MARCEL GROSSMANN MEETINGS, S. AC1

University of Rome, Italy

🔊: *The Formation of Two Component Advective Flows around Neutron Stars*

Jul 1 - 7, 2018

INTEGRAL SYMP., S. 4: ACCRETION AND EJECTION: GALACTIC AND EXTRAGALACTIC

INAF, Venice, Italy

🔊: *Is neutron star spectrum also an outcome of TCAF?*

Oct 15 - 20, 2017

INTEGRAL SYMP., S. 2: OUTBURSTING SOURCES: BHC, NS, AGN/BLAZARS

INAF, Venice, Italy

🔊: *Outburst of BHC H1743-322: Analysis of Spectral and Timing Properties Using TCAF Solution*

Oct 15 - 20, 2017

WIDE BAND SPECTRAL AND TIMING STUDIES OF COSMIC X-RAY SOURCES

TIFR Mumbai, India

🔊: *Is Neutron Star Spectrum also an Outcome of TCAF?*

January 10 - 13, 2017